

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE
PHYSICS

Paper 1

Tuesday 5 November 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- a protractor
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
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5	
6	
7	
8	
9	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

Stars form when enough dust and gas are pulled together in space.

0 1 . 1

What force causes the dust and gas to be pulled together?

[1 mark]

Tick (✓) **one** box.

Electrostatic force

☐

Frictional force

☐

Gravitational force

☐

Magnetic force

☐

0 1 . 2

Which **two** of the following are the main gases in a newly formed star?

[2 marks]

Tick (✓) **two** boxes.

Carbon dioxide

☐

Helium

☐

Hydrogen

☐

Nitrogen

☐

Oxygen

☐

0 1 . 3 Complete the sentence.

Choose the answer from the box.

[1 mark]

density	volume	wavelength
----------------	---------------	-------------------

The core of a star has a very high temperature and a very high _____.

The life cycle of a star is determined by the mass of the star.

0 1 . 4 What stage of the life cycle of a star is the Sun in?

[1 mark]

Tick (✓) **one** box.

Main sequence	☐
Protostar	☐
Red giant	☐
Red supergiant	☐

Question 1 continues on the next page

Turn over ►



0 1 . 5 Complete **Table 1** to show the stages of the life cycle of a star after the main sequence.

Tick (✓) **one** box in **each row** of **Table 1**.

The first two rows have been completed.

[2 marks]

Table 1

Stage of the life cycle	Stars about the same size as the Sun	Stars much bigger than the Sun
Red giant	✓	
Red supergiant		✓
White dwarf		
Black dwarf		
Neutron star		
Black hole		

0 1 . 6 There is a stage in the life cycle of some stars when elements are scattered throughout the universe.

What is the name of this stage?

[1 mark]



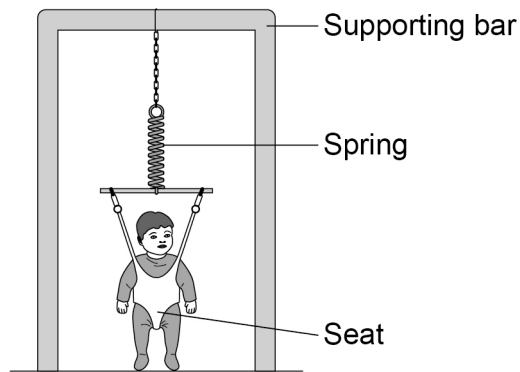
0 2

Figure 1 shows a baby in a baby bouncer.

The baby bouncer hangs from a spring and supports the weight of the baby.

When the baby jumps, the baby bounces up and down.

Figure 1



0 2 . 1

The mass of the baby is 7.5 kg.

gravitational field strength = 9.8 N/kg

Calculate the weight of the baby.

Use the equation:

$$\text{weight} = \text{mass} \times \text{gravitational field strength}$$

[2 marks]

Weight = _____ N

Question 2 continues on the next page

Turn over ►



An engineer is developing a new baby bouncer.

The engineer wants to test a new spring.

0 2 . 2

Describe how the engineer could determine the extension of the spring for a force of 150 N.

[3 marks]

0 2 . 3

Calculate the extension of the spring when the force applied to the spring is 150 N.

spring constant of spring = 400 N/m

Use the Physics Equations Sheet.

[3 marks]

Extension = _____ m



0 2 . 4

Calculate the elastic potential energy stored in the spring when the extension is 0.16 m.

spring constant of spring = 400 N/m

Use the equation:

$$\text{elastic potential energy} = \frac{1}{2} \times \text{spring constant} \times (\text{extension})^2$$

[2 marks]

Energy = _____ J

Question 2 continues on the next page

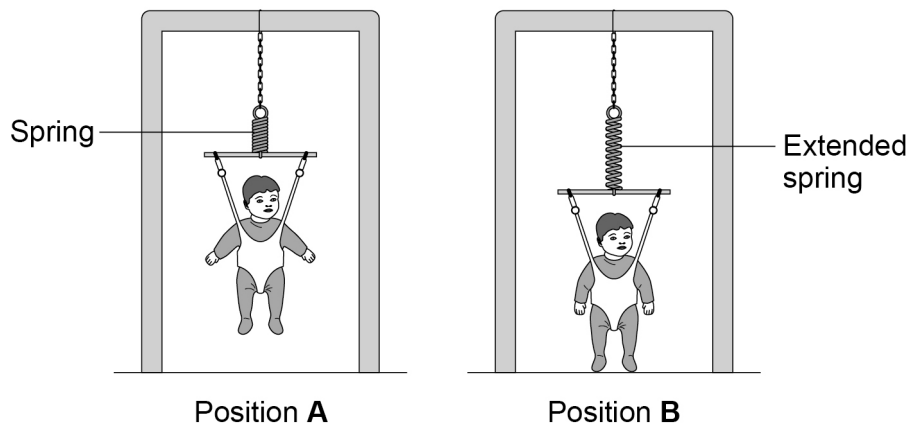
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0 2 . 5

Figure 2 shows the baby in two positions when using the baby bouncer.

Figure 2



At position **A** the baby is at maximum distance above the ground.

At position **B** the baby is touching the ground and the spring is extended.

Describe the changes to the energy stores as the baby moves from position **A** to position **B**.

[3 marks]

Turn over for the next question

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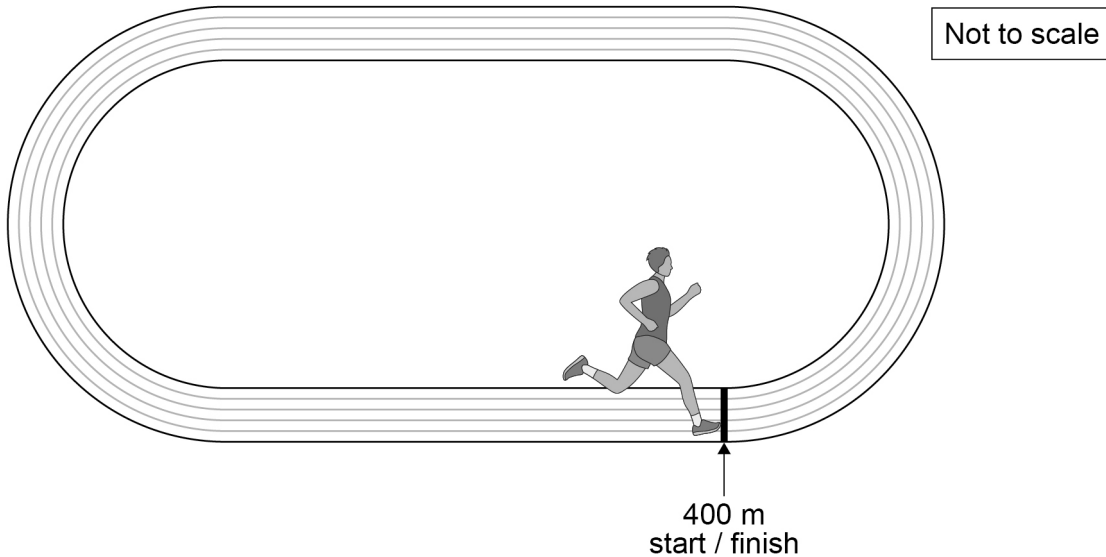


0 3

A student runs a race on a 400 m track.

Figure 3 shows the student at the end of the race.

Figure 3

**0 3****1**

The speed of the student is a scalar quantity.

A scalar quantity has magnitude only.

What is the difference between a scalar quantity and a vector quantity?

[1 mark]

0 3**2**

The student completed the 400 m race in a time of 80 s.

Calculate the average speed of the student.

Use the equation:

$$\text{average speed} = \frac{\text{distance}}{\text{time}}$$

[2 marks]

Average speed = _____ m/s



0 3 . 3

When the velocity of the student was 5.5 m/s the momentum of the student was 330 kg m/s.

Calculate the mass of the student.

Use the Physics Equations Sheet.

[3 marks]

Mass = _____ kg

0 3 . 4

The displacement of the student from the start of the race to the end of the race was zero.

Explain why.

Use **Figure 3**.

[2 marks]

0 3 . 5

At the end of the race the student had an acceleration of -1 m/s^2 .

What is meant by 'an acceleration of -1 m/s^2 '?

[2 marks]

Turn over for the next question



0 4

Nuclear fission reactions release energy in nuclear power stations.

Uranium is used as a fuel in nuclear power stations.

0 4 . 1

Which uranium isotope is used in nuclear power stations?

[1 mark]

Tick (✓) **one** box.

Uranium-235

☐

Uranium-238

☐

Uranium-239

☐**0 4 . 2**

Neutrons are released in nuclear fission reactions.

Give the relative mass and relative charge of a neutron.

[2 marks]

Relative mass _____

Relative charge _____



0 4 . 3

Figure 4 shows the nuclear fission of a uranium nucleus.

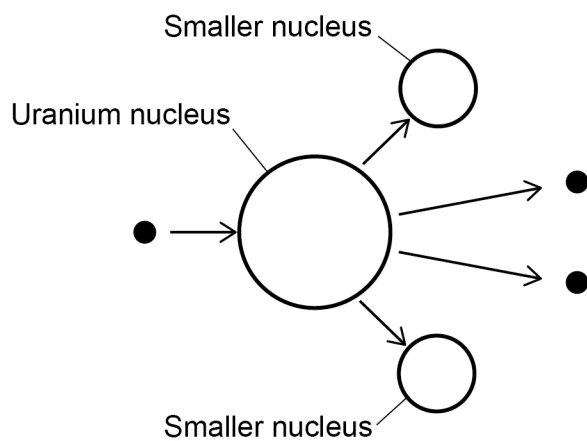
A neutron is absorbed by a uranium nucleus.

The nucleus then splits into two smaller nuclei and releases two neutrons.

Figure 4

Key

● Neutron



Complete **Figure 4** to show how a chain reaction could occur.

[2 marks]

Question 4 continues on the next page

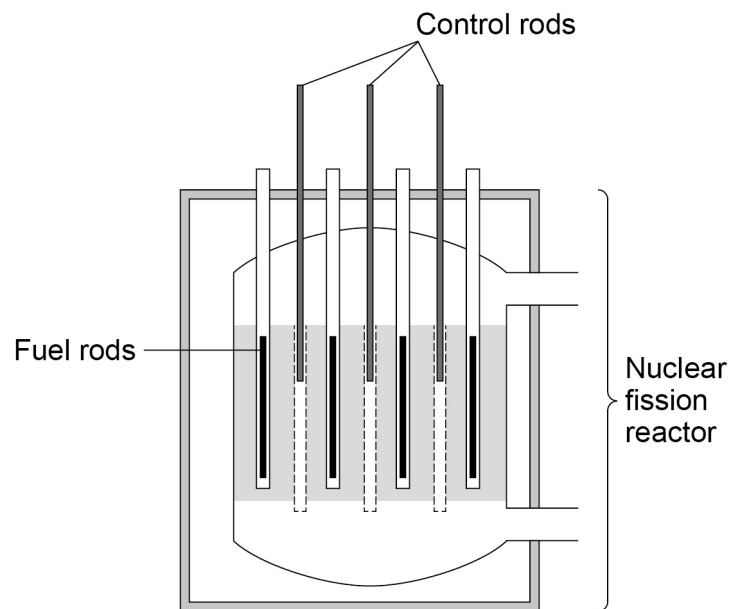
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Figure 5 shows the fuel rods and the control rods in a nuclear fission reactor.

The nuclear fission reactor is in a nuclear power station.

Figure 5



0 4 . 4

Describe how the control rods keep the power output of the nuclear fission reactor constant.

[3 marks]

0	4	.	5
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In an emergency, the nuclear fission reactions need to be stopped.

Suggest how the control rods could be used to stop the nuclear fission reactions.

Use **Figure 5**.

[1 mark]

Question 4 continues on the next page



0 4 . 6

Coal and oil are also used as fuel in power stations.

Table 2 shows the energy released when 1 kg of fuel is used to generate electricity in each power station.

Table 2

Fuel	Energy in kWh
Coal	18
Oil	12
Uranium	24 000 000

Compare the use of these different fuels for generating electricity.

You should use information from **Table 2** and your own knowledge.

[4 marks]

Extra space _____



0 5

A person hangs some wet clothes outside to dry.

0 5 . 1

What process causes the wet clothes to dry?

[1 mark]Tick (✓) **one** box.

Boiling

☐

Condensation

☐

Evaporation

☐

Freezing

☐**0 5 . 2**Give **two** factors that would affect the rate of drying of the clothes outside.**[2 marks]**

1 _____

2 _____

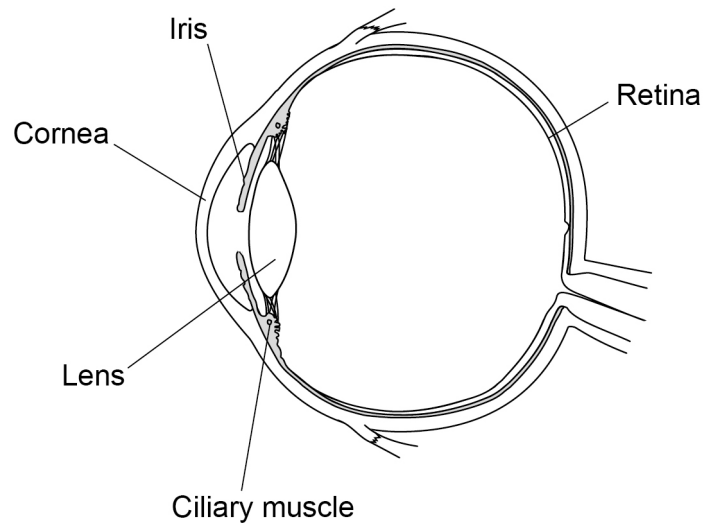
Question 5 continues on the next page**Turn over ►**

Table 3 gives information about two electrical devices used to dry clothes.

	Power rating in W	Time taken to dry clothes in hours	Cost of device in \$
Device A	3000	0.75	250
Device B	150	6	510

[6 marks]

[illegible]

0 6**Figure 6** shows a human eye.**Figure 6****0 6 . 1**Which part of the eye in **Figure 6** controls the amount of light entering the eye?**[1 mark]**

0 6 . 2

Refraction occurs as light passes through the lens.

Where else in the eye in **Figure 6** does the refraction of light occur?**[1 mark]**

Question 6 continues on the next page**Turn over ►**

0 6 . 3

The refractive index of a medium is defined as

$$\frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}}$$

The refractive index of the medium of a lens is 1.44

speed of light in a vacuum = 300 000 000 m/s

Calculate the speed of light in the lens.

Give your answer in standard form.

[4 marks]

Speed (standard form) = _____ m/s



0 6 . 4

A person looks at a tree.

The height of the image of the tree on the retina of the person's eye is 5.5 mm.

The magnification of the eye is 0.0012

Calculate the height of the tree.

Give your answer to 2 significant figures.

[4 marks]

Height of tree (2 significant figures) = _____ m

10

Turn over for the next question**Turn over ►**

0 7

Electromagnetic waves have many practical uses.

0 7 . 1

Table 4 shows the uses of some parts of the electromagnetic spectrum.

Table 4

Use	Electromagnetic wave
Bluetooth communication	Radio waves
Medical imaging	X-rays
Night vision devices	
Security marking	Ultraviolet
Sterilising surgical equipment	

Complete **Table 4**.

[2 marks]

0 7 . 2

X-rays are used to produce X-ray images of bones inside human bodies.

A doctor uses an X-ray machine to produce hundreds of X-ray images each year.

The doctor stands behind a lead screen whilst the X-ray images are produced.

Explain why it is important for the doctor to stand behind the lead screen.

[3 marks]



0 7 . 3 **Figure 7** shows an X-ray image of a person with a metal artificial hip.

The hip is where a person's leg bone joins the body.

The muscle labelled in **Figure 7** is healthy tissue.

Figure 7



Different substances inside a body appear different on an X-ray image.

Explain why X-rays can produce the image of the different substances in **Figure 7**.

You should refer to the parts labelled in **Figure 7**.

[3 marks]

Turn over for the next question



0 8

The mains electrical power supply in the UK is an alternating current (ac) supply.
Batteries supply a direct current (dc).

0 8 . 1

Describe the difference between an alternating current (ac) and a direct current (dc).

[2 marks]

A mains electrical fan has a metal casing.

The electrical fan has an earth wire and fuse.

0 8 . 2

The current in the electrical fan is 0.17 A.

Which fuse would be safest to use with this electrical fan?

[1 mark]

Tick (✓) **one** box.

1 A

☐

5 A

☐

13 A

☐

0 8 . 3

An electrical fault causes the metal casing of the fan to become live.

Explain how the earth wire and fuse prevent a person using the electrical fan from getting an electric shock when the fault occurs.

[3 marks]

0 8 . 4

Explain why some mains electrical devices do **not** need an earth wire.

[2 marks]

8

Turn over for the next question

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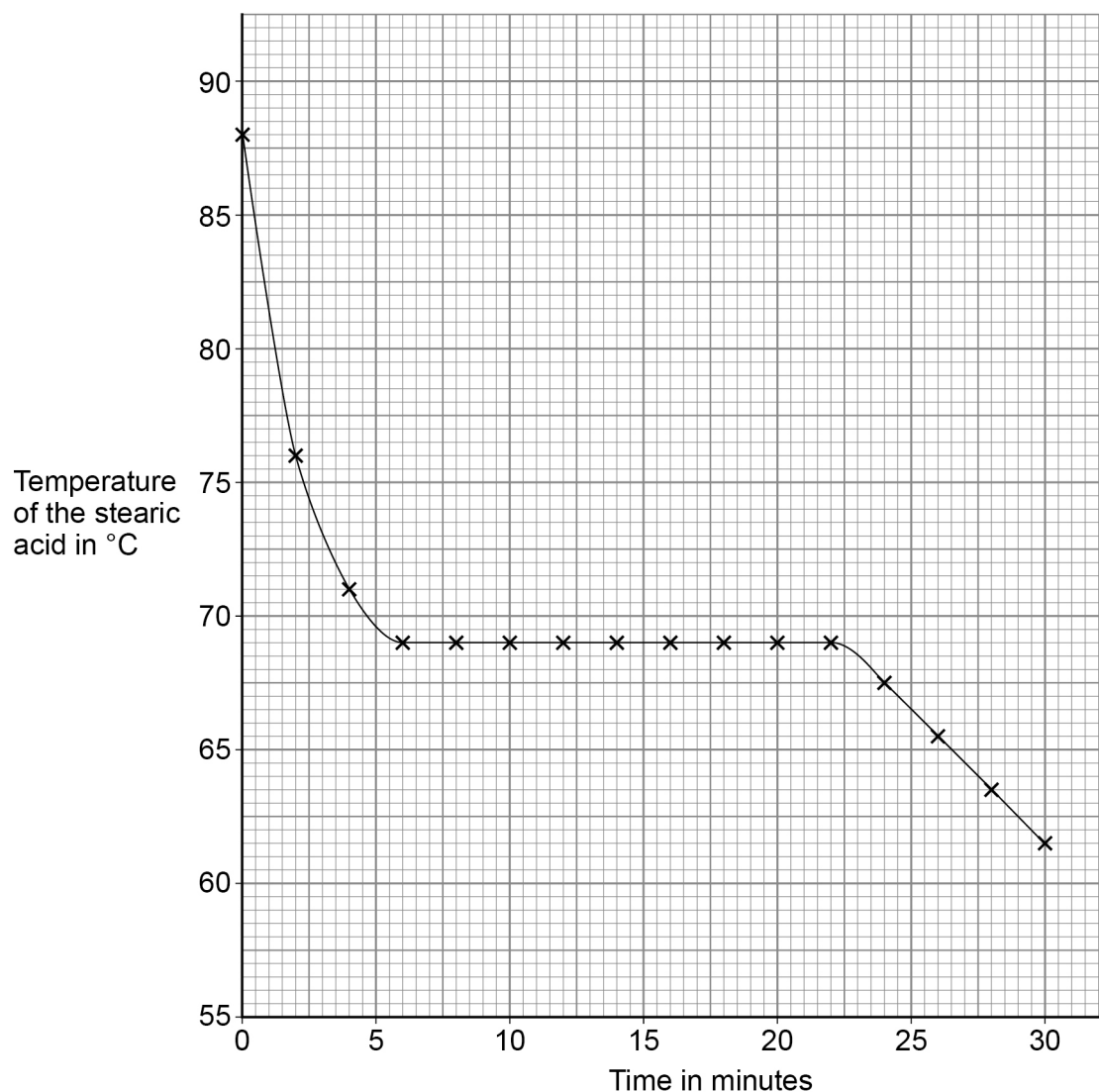
0 9

A student investigated the change of state of stearic acid as it cooled.

The student measured the temperature of the stearic acid for 30 minutes.

Figure 8 shows the results.

Figure 8



0 9 . 1

What was the time between each temperature reading in the investigation?

Use **Figure 8**.

[1 mark]

Time = _____ minutes



The stearic acid changed state from a liquid to a solid during the investigation.

0 9 . 2

Determine the temperature change of the stearic acid when in the liquid state.

Use **Figure 8**.

[2 marks]

Temperature change = _____ °C

0 9 . 3

The average rate of energy transfer during the change of state was 2.4 W.

specific latent heat of fusion of stearic acid = 2.0×10^5 J/kg

Calculate the mass of stearic acid the student used in the investigation.

Use data from **Figure 8**.

[6 marks]

Mass = _____ kg

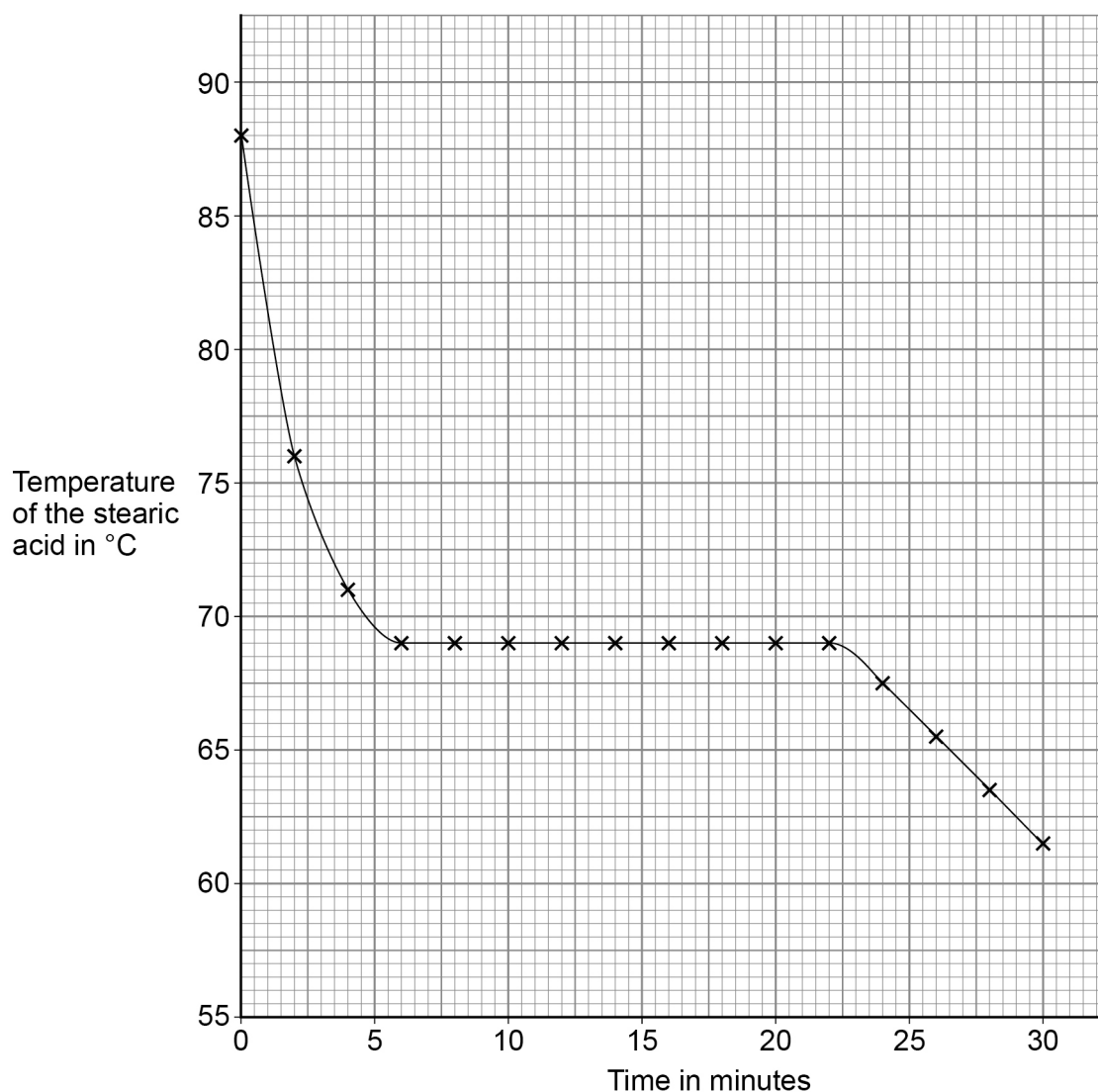
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0 9 . 4 Figure 8 is repeated below.

Figure 8



The student repeated the investigation on a second day.

The student used the same mass of stearic acid.

The starting temperature of the stearic acid was the same on both days.

The room temperature was higher on the second day.

Draw a line on **Figure 8**, above, to show how the liquid stearic acid cooled and changed state on the second day.

[2 marks]

11

END OF QUESTIONS



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